Lab Work Assignment 4: MLflow Experiment Tracking and Artifact Management

Course: ML Operations

Objective:
The objective of this lab is to set up an MLflow tracking server, log all relevant artifacts (such as configurations, parameters, metrics, and files), and demonstrate effective experiment management. Students are expected to integrate MLflow into their machine learning workflows to enhance reproducibility, track experiments, and log essential information for model training and evaluation.

Resources:

- MLflow Documentation
- MLflow GitHub Repository
- Example MLflow Project

Submission Deadline: 21 October

Part 1: MLflow Tracking Server Setup

Tracking Server Initialization

Task:

- Install MLflow and set up a local or remote MLflow tracking server.
- Demonstrate that the server is correctly running and accessible through the MLflow UI.
- Ensure that you include a `.gitignore` file to exclude unnecessary files from version control.

GitHub Repository Setup

Task:

- Use GitHub (or any similar platform) to create a repository for your project.
- Commit all necessary code and configurations to the repository.

Part 2: Logging Parameters, Configs, and Metrics in MLflow

Logging Parameters and Configurations

Task:

- Create a configuration file (e.g., `config.yaml`) to manage important model and training parameters, such as learning rates, batch sizes, and data paths.
- Log these parameters in MLflow for each experiment run.

Metrics Logging

Task:

- Log key metrics, such as accuracy, loss, or any relevant evaluation scores, to MLflow.
- Define an appropriate interval for logging metrics (e.g., per epoch or after every n iterations). Ensure that metrics are visualized in the MLflow UI.

Part 3: Logging Output Artifacts

Model Artifact Logging

Task:

- Ensure that your best-trained model is logged as an artifact in MLflow. This includes saving model weights and any associated files required for inference.
- Demonstrate that the logged model can be downloaded and used for further evaluation or deployment.

Part 4: Experimentation with MLflow

Experiment Creation and Run Management

Task:

- Create an experiment in MLflow and log multiple runs under this experiment. Each run should correspond to different configurations or parameter values, demonstrating experimentation and tuning.
- Compare the logged runs in the MLflow UI to analyze model performance.

Run Name Hierarchy (Additional Points)

Task *(Optional, for additional points)*:

- Create a hierarchical naming convention for runs, representing different stages or configurations (e.g., “Experiment 1 - Stage 1: Data Preprocessing”, “Experiment 1 - Stage 2: Model Training”).
- Demonstrate how the hierarchy improves experiment tracking and interpretation within MLflow.

Part 5: Lab Report

Content:

- **Introduction**: Explain the importance of experiment tracking in machine learning workflows and introduce MLflow as a tool for this purpose.
- **Tracking Server Setup**: Describe how you set up the MLflow tracking server and the process of integrating it into your workflow.
- **Logging Details**: Provide an overview of the parameters, metrics, and artifacts you logged during the experiment.
- **Experimentation Process**: Discuss how you managed multiple runs within the same experiment and how the logged information helped with model evaluation.
- **Reflection**: Reflect on the benefits and challenges of using MLflow in your project, including any additional improvements that could be made.

Grading Criteria

- **MLflow Server Setup**: Proper setup and demonstration of the MLflow tracking server.
- **Experiment Logging**: Successful logging of parameters, metrics, and artifacts, as well as appropriate use of the MLflow UI to track and compare runs.
- **Version Control**: Effective use of GitHub (or similar) for code management, including clear commit messages and an organized repository.
- **Lab Report**: Clarity, depth, and completeness of the lab report, including the reflection on the benefits and challenges of MLflow.